FIRE ALARM SYSTEM ARCHITECTURE & INTEGRATION:
1. Central Networked Fire Alarm Control Panel (FACP) with 8 Class-A Signaling Line Circuits (SLC) loops per EN 54 & NBC 2016.
2. Addressable multi-sensor optical smoke and rate-of-rise heat detectors installed in all office zones, lobbies, and electrical shafts.
3. Addressable manual pull stations (MCP) with transparent protective flip-covers positioned at every emergency exit doorway.
4. Audio-visual strobe sounders delivering 100 dBA audio output with synchronized 75 candela xenon flashes.
5. Interlock control modules wired for automatic tripping of AHUs, closing of fire smoke dampers, and grounding of elevators.

Loop ID	Floor / Building Zone	Detectors	Manual Calls	Control Modules	Total Dev
Loop 1	Ground Floor North Wing & Lobby	84 Smoke / 12 Heat	8 MCPs	14 Modules	118 Devices
Loop 2	Ground Floor South Wing & Portico	78 Smoke / 10 Heat	8 MCPs	12 Modules	108 Devices
Loop 3	First Floor Legislative Chambers	96 Smoke / 16 Heat	12 MCPs	18 Modules	142 Devices
Loop 4	Second Floor Cabinet & Ministers	90 Smoke / 14 Heat	10 MCPs	16 Modules	130 Devices
Loop 5	Basement Parking & Plant Rooms	65 Heat / 20 Flame	12 MCPs	22 Modules	119 Devices

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: INTELLIGENT ADDRESSABLE FIRE ALARM SYSTEM LOO		
DRAWING NO: SL-288-Amaravati-LA-FIRE-AL-01	REV: A1	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	